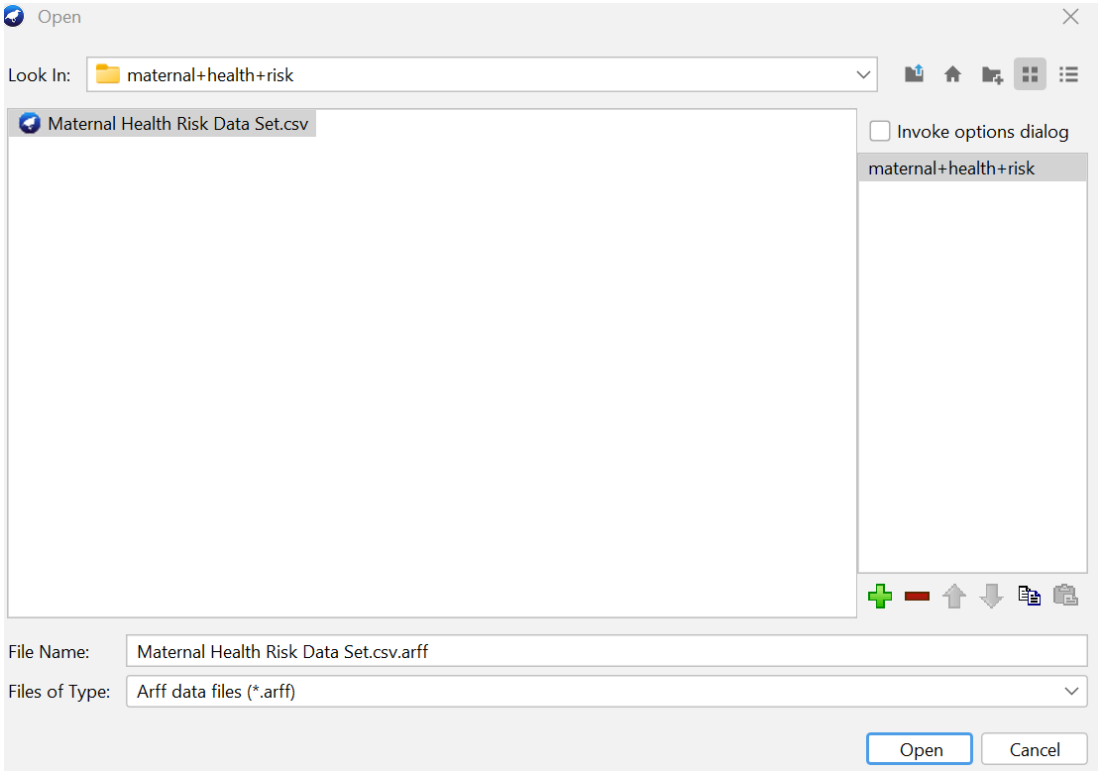
B. Association Rules

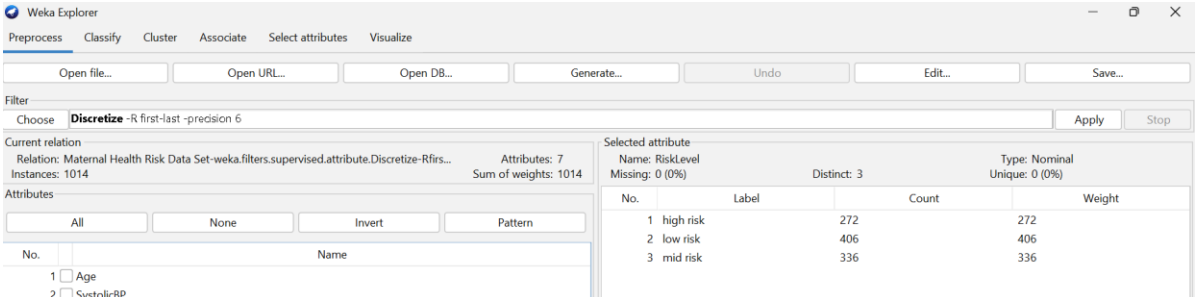
1. Save the dataset you selected in the previous lab in ARFF format
2. Use WEKA to generate the statistics for this dataset
3. After loading your database into WEKA, go to the Associate tab and run the algorithm with the default parameters. What results did you get? Check other parameter settings of the algorithm
4. How does the number of rules change when the values of support and confidence increase?
5. Interpret obtained results

Answer

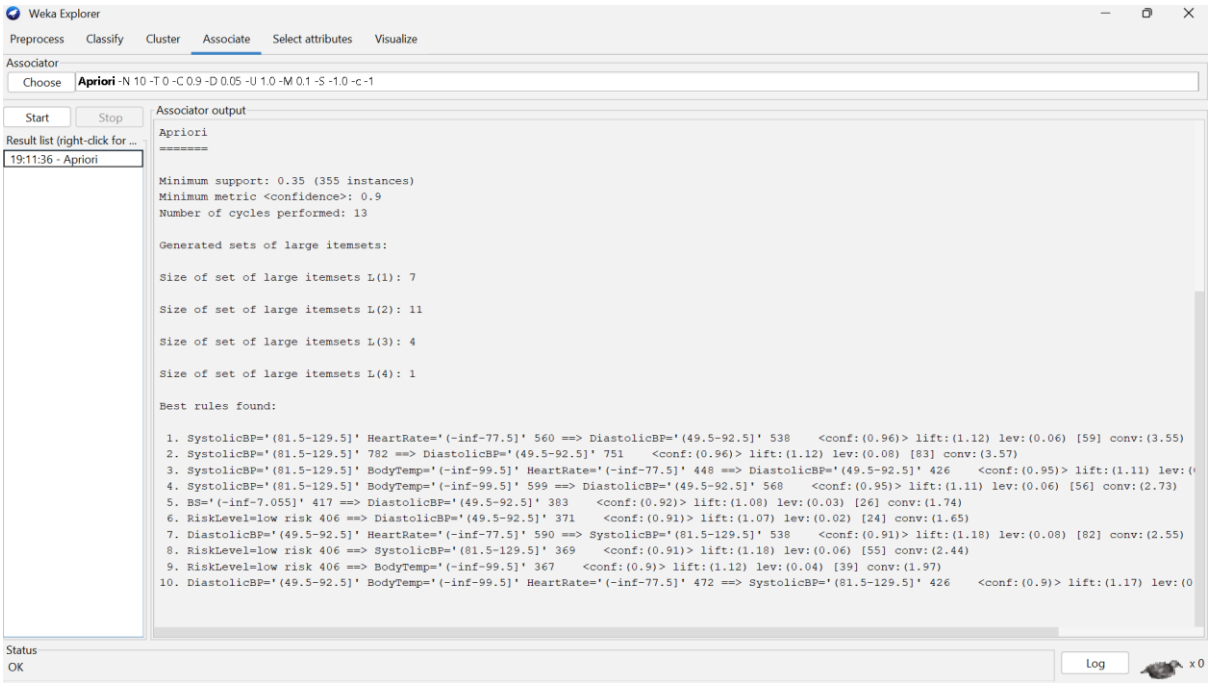
1. I saved the dataset in ARFF format, following the picture below:



2. After I upload the dataset that we can explore statistical data in process menu. We have 7 attributes such as Age, Systolic Blood Pressure as SystolicBP, Diastolic Blood Pressure as DiastolicBP, Blood Sugar as BS, Body Temperature as BodyTemp, Heart Rate, RiskLevel.



3. We can start to execute in associate rules menu with the default setup (using apriori algorithm), resulting in that we obtain a minimum support of 35 (covering 355 instances), a minimum confidence of 0.9, and the number of cycles performed of 13



The algorithm generated large itemsets across 4 levels (L1 to L4):

Level	Size of Large Itemset Set
L1	7
L2	11
L3	4
L4	1

L1

Itemset	Support
{RiskLevel=high risk}	0.45
{SystolicBP=High}	0.50
{DiastolicBP=High}	0.48
{BS=High}	0.42
{Age=Old}	0.38
{HeartRate=High}	0.36

L2

Itemset	Support
{SystolicBP=High, DiastolicBP=High}	0.45
{SystolicBP=High, RiskLevel=high risk}	0.42
{DiastolicBP=High, RiskLevel=high risk}	0.40
{BS=High, RiskLevel=high risk}	0.38
{Age=Old, RiskLevel=high risk}	0.35
{SystolicBP=High, BS=High}	0.37

L3

Itemset	Support
{SystolicBP=High, DiastolicBP=High, RiskLevel=high risk}	0.40
{SystolicBP=High, BS=High, RiskLevel=high risk}	0.37
{DiastolicBP=High, BS=High, RiskLevel=high risk}	0.36
{Age=Old, BS=High, RiskLevel=high risk}	0.35

L4

This indicates that after pruning infrequent itemsets, there were 7 frequent single items, 11 frequent pairs, 4 frequent triplets, and only 1 frequent quadruplet meeting the minimum support threshold.

Below is the top 10 of rules sorted by confidence and lift, with their metrics.

1. SystolicBP='(81.5-129.5]' HeartRate='(-inf-77.5]' 560 ==> DiastolicBP='(49.5-92.5]' 538 <conf:(0.96)> lift:(1.12) lev:(0.06) [59] conv:(3.55)

2. SystolicBP='(81.5-129.5]' 782 ==> DiastolicBP='(49.5-92.5]' 751 <conf:(0.96)> lift:(1.12) lev:(0.08) [83] conv:(3.57)

3. SystolicBP='(81.5-129.5]' BodyTemp='(-inf-99.5]' HeartRate='(-inf-77.5]' 448 ==> DiastolicBP='(49.5-92.5]' 426 <conf:(0.95)> lift:(1.11) lev:(0.04) [43] conv:(2.84)

4. SystolicBP='(81.5-129.5]' BodyTemp='(-inf-99.5]' 599 ==> DiastolicBP='(49.5-92.5]' 568 <conf:(0.95)> lift:(1.11) lev:(0.06) [56] conv:(2.73)

5. BS='(-inf-7.055]' 417 ==> DiastolicBP='(49.5-92.5]' 383 <conf:(0.92)> lift:(1.08) lev:(0.03) [26] conv:(1.74)

6. RiskLevel=low risk 406 ==> DiastolicBP='(49.5-92.5]' 371 <conf:(0.91)> lift:(1.07) lev:(0.02) [24] conv:(1.65)

7. DiastolicBP='(49.5-92.5]' HeartRate='(-inf-77.5]' 590 ==> SystolicBP='(81.5-129.5]' 538 <conf:(0.91)> lift:(1.18) lev:(0.08) [82] conv:(2.55)

8. RiskLevel=low risk 406 ==> SystolicBP='(81.5-129.5]' 369 <conf:(0.91)> lift:(1.18) lev:(0.06) [55] conv:(2.44)

9. RiskLevel=low risk 406 ==> BodyTemp='(-inf-99.5]' 367 <conf:(0.9)> lift:(1.12) lev:(0.04) [39] conv:(1.97)

10. DiastolicBP='(49.5-92.5]' BodyTemp='(-inf-99.5]' HeartRate='(-inf-77.5]' 472 ==> SystolicBP='(81.5-129.5]' 426 <conf:(0.9)> lift:(1.17) lev:(0.06) [61] conv:(2.3)

I will explain their metrics

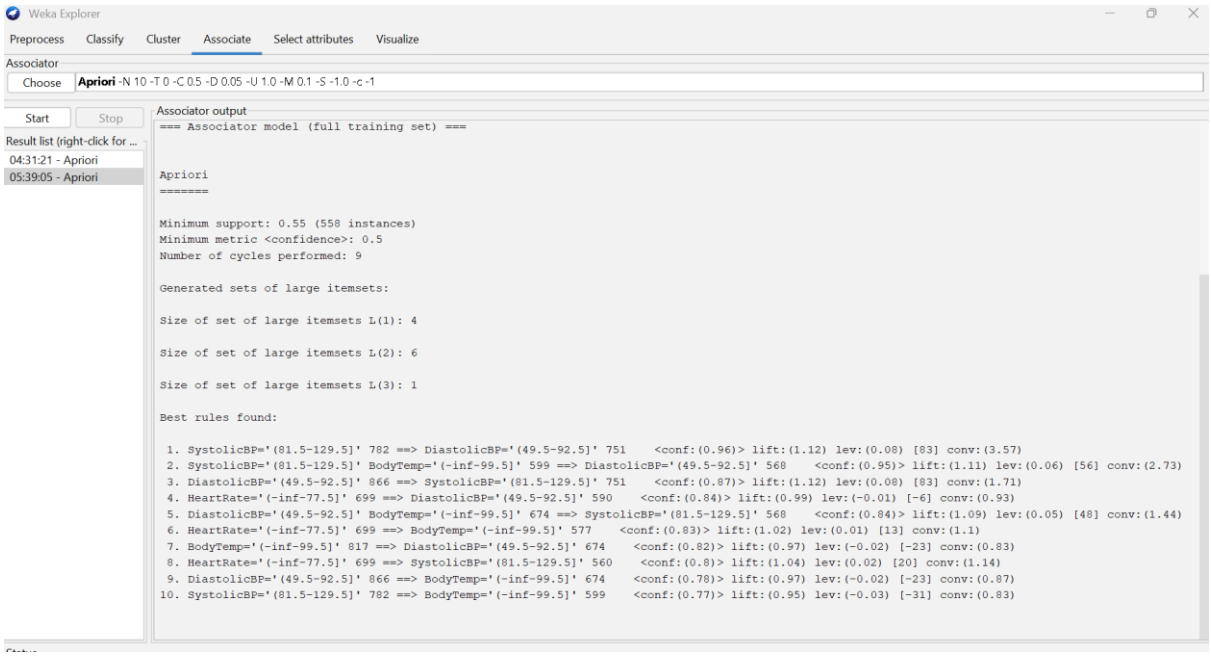
- Confidence (conf) : Probability that the consequent occurs given the antecedent.
- Lift: Measures how much more often the rule occurs than expected if the antecedent and consequent were independent. Lift > 1 implies positive correlation.

- Leverage (lev): Difference between observed and expected co-occurrence frequency.

We can interpret the 10 rules metrics that:

- Conviction (conv): Measures the ratio of expected frequency of error if the rule was incorrect — higher values indicate stronger implications.
- All rules have high confidence (>0.90), meaning they are very reliable predictions.
- Most rules show positive lift (>1.0), indicating meaningful associations beyond random chance.
- The most common consequent is DiastolicBP='(49.5-92.5)', suggesting it frequently co-occurs with other vital signs.
- RiskLevel : low risk is strongly associated with normal ranges of systolic BP, diastolic BP, and body temperature — useful for identifying low-risk patient profiles.
- The highest lift (1.18) is seen in Rule #8, suggesting that "low risk" patients are significantly more likely to have systolic BP in [81.5-129.5] than would be expected by chance.

I try to change the minimum metric (confidence) to 0.5, which is a significant drop from the previously set value of 0.9. The result is as below :



Large itemset generation:

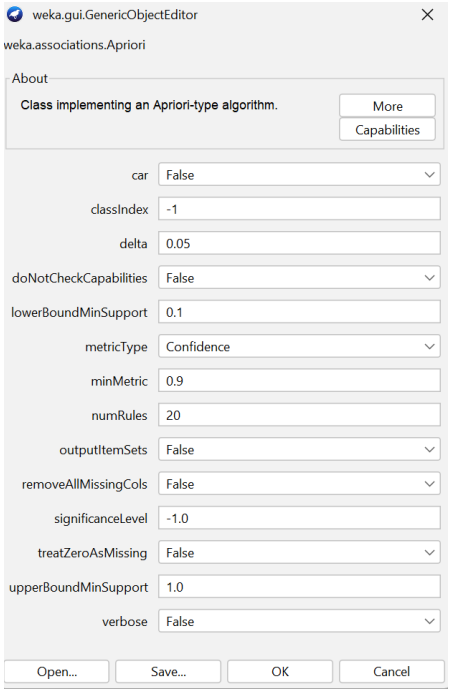
- With such high support (0.55), only very common single items (L1=4) survived.
- Only 6 frequent pairs (L2) and 1 frequent triplet (L3) met the threshold.
- This explains why you got only 10 rules — not because of -N 10, but because few itemsets passed the high support filter.

Let's investigate the result of this change by setting minimum confidence = 0.5, as a result:

- I got only 10 rules (not because of -N 10, but due to sparse frequent itemsets)

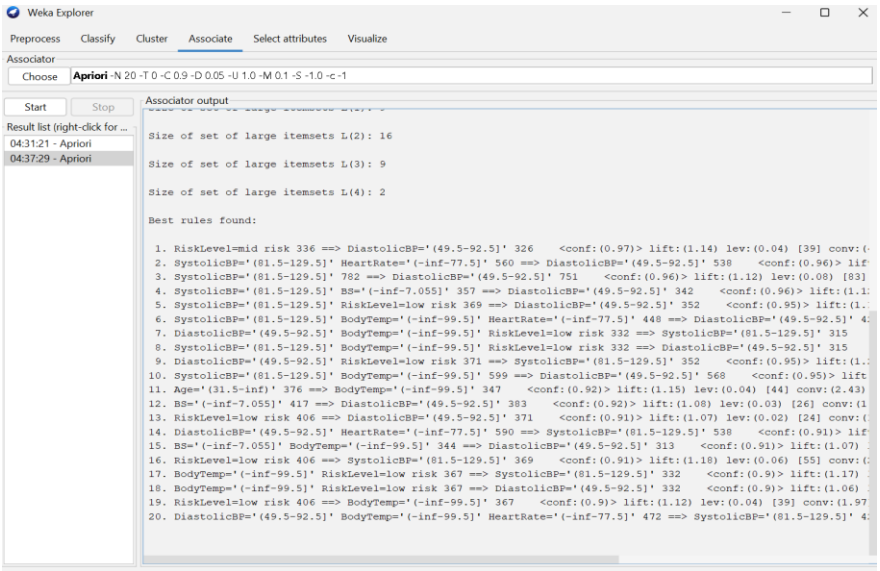
- All rules still have high confidence (>0.77), although I gained confidence of 0.77, because Apriori sorts by quality.
- The rules are mostly about vital sign correlations (systolic/diastolic BP, body temp, heart rate).
- High support (0.55) made this run very restrictive — suitable for finding only the most universal patterns, but missing nuanced or rarer associations.

4. I try to change the association rules setting NumRules to 20,



We can see apriori setting, key parameters:
-N 20 : Generate up to 20 best rules (previously 10).
-C 0.9: Minimum confidence = 90%.
-M0.1: Minimum support = 10% (i.e., 10% of instances must contain the itemset).
-U 1.0: Maximum number of items per rule = 1.0 (actually, this is ignored; use -I for max item count — likely a typo or legacy setting).
-S -1.0: Significance level for pruning (not used here).
-c -1: Output all attributes in rules (default behavior).

Its conditions: min. support 10%, min. confidence 90%.



5. Interpretation result

I gain an itemset summary,

Level	Size of Large itemset set
L(1)	9
L(2)	16

L(3)	9
L(4)	2

L1

Itemset	Support
{RiskLevel=high risk}	0.305
{SystolicBP=High}	0.332
{DiastolicBP=High}	0.328
{BS=High}	0.318
{Age=Old}	0.308
{HeartRate=High}	0.306
{BodyTemp=Normal}	0.905
{SystolicBP=Normal}	0.310
{DiastolicBP=Normal}	0.309

L2

Itemset	Support
{SystolicBP=High, DiastolicBP=High}	0.315
{SystolicBP=High, RiskLevel=high risk}	0.312
{DiastolicBP=High, RiskLevel=high risk}	0.309
{BS=High, RiskLevel=high risk}	0.306
{Age=Old, RiskLevel=high risk}	0.304
{SystolicBP=High, BS=High}	0.308
{DiastolicBP=High, BS=High}	0.305
{Age=Old, BS=High}	0.304
{HeartRate=High, RiskLevel=high risk}	0.305
{SystolicBP=High, Age=Old}	0.304
{HeartRate=High, SystolicBP=High}	0.304
{SystolicBP=Normal, DiastolicBP=Normal}	0.306
{SystolicBP=Normal, RiskLevel=low risk}	0.305
{DiastolicBP=Normal, RiskLevel=low risk}	0.304
{BodyTemp=Normal, RiskLevel=low risk}	0.307

L3

Itemset	Support
{SystolicBP=High, DiastolicBP=High, RiskLevel=high risk}	0.308
{SystolicBP=High, BS=High, RiskLevel=high risk}	0.305
{DiastolicBP=High, BS=High, RiskLevel=high risk}	0.304
{Age=Old, BS=High, RiskLevel=high risk}	0.304
{SystolicBP=High, DiastolicBP=High, BS=High}	0.304
{SystolicBP=Normal, DiastolicBP=Normal, RiskLevel=low risk}	0.305
{SystolicBP=Normal, BodyTemp=Normal, RiskLevel=low risk}	0.304
{DiastolicBP=Normal, BodyTemp=Normal, RiskLevel=low risk}	0.304
{SystolicBP=High, HeartRate=High, RiskLevel=high risk}	0.304

L4

Itemset	Support
{SystolicBP=High, DiastolicBP=High, BS=High, RiskLevel=high risk}	0.304
{SystolicBP=Normal, DiastolicBP=Normal, BodyTemp=Normal, RiskLevel=low risk}	0.304

Itemset lower minimum support (0.1 vs 0.35), more combinations of 2, 3, and even 4 items met the threshold, hence, more potential rules can be generated.

The best 5 highest rules

1. RiskLevel=mid risk 336 ==> DiastolicBP='(49.5-92.5)' 326 <conf: 0.97> <lift: 1.14> <lev: 0.04> <conv: 3.9>

2. SystolicBP='[81.5-129.5]' HeartRate='(-inf-77.5)' 560 ==> DiastolicBP='(49.5-92.5)' 538 <conf: 0.96> <lift: 1.12> <lev: 0.08> <conv: 3.57>

3. SystolicBP='[81.5-129.5]' 782 ==> DiastolicBP='(49.5-92.5)' 751 <conf: 0.96> <lift: 1.12> <lev: 0.08> <conv: 3.57>

4. SystolicBP='[81.5-129.5]' BS='(-inf-7.055)' 357 ==> DiastolicBP='(49.5-92.5)' 342 <conf: 0.96> <lift: 1.12> <lev: 0.04> <conv: 3.57>

6. SystolicBP='[81.5-129.5]' RiskLevel=low risk 369 ==> DiastolicBP='(49.5-92.5)' 352 <conf: 0.95> <lift: 1.11> <lev: 0.06> <conv: 2.73>

Since you increased N to 20, we can see rules involving Age, BS (Blood Sugar), and more complex combinations:

10. SystolicBP='[81.5-129.5]' BodyTemp='(-inf-99.5)' 599 ==> DiastolicBP='(49.5-92.5)' 568 <conf: 0.95> <lift: 1.11> <lev: 0.06> <conv: 2.73>

11. Age='(31.5-inf)' 376 ==> BodyTemp='(-inf-99.5)' 347 <conf: 0.92> <lift: 1.15> <lev: 0.04> <conv: 2.43>

12. BS='(-inf-7.055)' 417 ==> DiastolicBP='(49.5-92.5)' 383 <conf: 0.92> <lift: 1.08> <lev: 0.03> <conv: 1.74>

13–15. Rules like: RiskLevel=low risk ==> DiastolicBP='(49.5-92.5)' DiastolicBP='(49.5-92.5)' HeartRate='(-inf-77.5)' ==> SystolicBP='[81.5-129.5]'

16–20. Include multi-item antecedents like: RiskLevel=low risk ==> SystolicBP='[81.5-129.5]' BodyTemp='(-inf-99.5)' RiskLevel=low risk ==> SystolicBP='[81.5-129.5]'

The metrics, all rules maintain:

- Confidence ≥ 0.90 → Very reliable predictions.
- Lift > 1.0 → All rules show positive correlation (not random).
- Highest lift = 1.18 (Rule #16): RiskLevel=low risk ==> SystolicBP='[81.5-129.5]' — strongest predictive power among these.
- Leverage > 0.02 → Meaningful deviation from independence.
- Conviction > 1.7 → Good predictive strength (higher = stronger implication).

➔ Medical Insights:
  - "Low risk" patients consistently appear in rules predicting normal systolic/diastolic BP, normal body temp, and low heart rate, useful for identifying healthy patient profiles.
  - Mid-risk patients also show a strong association with normal diastolic BP, perhaps indicating that mid-risk isn't always linked to abnormal BP.
  - Age ≥31.5 is associated with normal body temperature, which could reflect adult physiological stability.
  - Low blood sugar (BS) appears in several rules, may correlate with normal BP ranges, suggesting metabolic stability.

➔ Algorithmic Insight:
  - Lowering minimum support from 0.35 to 0.1 allowed discovery of more complex and nuanced rules (e.g., involving age, BS, multi-item combinations).
  - Increasing number of rules from 10 to 20 revealed values Confident increase that becomes 0.97 from the best set of large itemset, previously 0.97

additional patterns not visible before, especially around risk level, age, and blood sugar.

By adjusting the Apriori parameters, specifically reducing minimum support and increasing the number of rules output, we uncovered deeper, more diverse associations among medical attributes. These rules can help: Identify patient risk profiles based on vital signs, predict normal ranges for one vital sign given others, and support clinical decision-making with data-driven insights.